

BIOMASS ENERGY & AI — Q&A: Crew, Financial Model, Entity Structure, Governance

Answers to four questions raised 2026-07-09, drawing on the forest workforce plan, EVM plan, business case, stakeholders doc, and NGO-setup materials.

1. How big is the crew now, and how does it scale up?

The forestry crew belongs to 御杖村森林組合 (Mitsue Kanko), not the project entity — the project's role is to fund its growth, not to hire directly.

- **Current size:** very small — sources disagree slightly (3 regular staff avg. age ~33 per one recruiting interview; "2 field + 2 office" = 4 total per another listing). Either way, a 3–4 person operation.
- **Current output:** ~66–68 ha/yr thinning, only ~120–300 m³/yr recovered timber — tiny relative to the village's 7,051 ha of forest, because it's subsidized thinning, not mechanized harvest.
- **Scaling path (the plan's core mechanism):** energy revenue pays for more crew + machines (harvester/forwarder, chipper, drying) → bigger crew harvests more → feeds a bigger biomass plant — a self-funding loop modeled on Shimokawa/Nishiawakura.
 - Doubled crew → ~140 ha/yr → 50% of forest over 25 yr → ~1.15 MWe (**recommended baseline**)
 - Tripled crew, full mechanization → ~200 ha/yr → ~1.6 MWe
- **Where new people come from:** partly the co-op's own hiring, but the plan names a concrete external channel — 天川村森林塾 (Tenkawa Forest School), an 8-day felling/chainsaw course that has trained 60+ people since 2017 explicitly "from inside and outside the village." That's the up-skilling pipeline for the doubled crew.

2. High-level financial model

Two budget tiers exist — kept separate, not on the same books.

Phase 0–3 build (¥220M BAC / ¥245M total, Baseline Rev 1, May 2026)

Item	¥M
School renovation	38
Solar PV ~100kW	22
Battery storage	12
EV charging (4 stations)	15
Data center fitout	20
Forestry (5–10ha + roads + replant)	25
Fiber connectivity	10
Testing/commissioning	8
Contingency (18%)	27
Phase 3 total	177
Management reserve	25
Total project budget	245

Item	¥M
Funding target (unsecured; ¥0 raised/committed to date)	192 (gap if fully realized: ¥28–53M)

Phase 4 forward capital (outside the PMB, funded by grants + operating revenue)

Item	¥M
CHP fleet (2×0.6 MWe)	~800
Fuel-prep (drying/chipping/handling)	~120 (range 75–215)

Illustrative operating P&L by Year 5 (Base case)

	Low	Base	High
Annual revenue (6 streams: FIT electricity, heat, DC hosting, EV, broadleaf/J-Credit, other)	¥28M	¥45M	¥67M
Operating cost (~60%)	¥18M	¥27M	¥35M
Net surplus	¥10M	¥18M	¥32M
Payback on ¥200–290M capital	~18 yr	~14 yr	~10 yr

The whole build is gated (G1–G4); the village/investors never commit past what's funded and de-risked at each stage.

3. Is an NPO the right vehicle, or would a social enterprise (K.K. with capped/rededicated profit in the articles) be more suitable?

Already decided, and more specific than a straight NPO-or-KK choice: a **two-tier entity**.

- **Tier 1 — 一般社団法人 (Ippan Shadan Hojin, non-profit type)**, with a path toward NPO法人/公益認定 later (per the stakeholders doc and lawyer-meeting prep with SIIF's Yuya Kato). Holds the charter, the village 官民連携 relationship, IP, forestry programme, J-Credit registration. Receives grants/donations — legally cannot distribute profit. The draft teikan already encodes this: **Article 26 — "no distribution of surplus."**
- **Tier 2 — 合同会社 (GK, a for-profit LLC-type entity, not a K.K.)**. Holds the actual assets — DC, biomass CHP, solar, battery, FIT/hosting contracts. This is where equity investment and dividends legally live.
- **The link:** Tier 1 owns the majority of the Tier 2 GK plus a charter "mission-lock" clause, so investors get a real but minority, contracted return and can't redirect the mission or strip community assets.

The "social enterprise with capped/rededicated profit written into the articles" is essentially **what the ISH already is under Japanese law** — non-distribution is a statutory requirement of the non-profit-type ISH, not something that needs inventing via a KK's articles. A KK doing the same thing would work against the grain of its own legal form and forfeit grant/donation eligibility. Given the funding stack leans heavily on 交付金, foundation grants, and MoE subsidy (all of which require or strongly favor a non-profit recipient), the two-tier ISH+GK structure fits — a KK would only make sense if the project needed equity as its *primary* capital source, which it doesn't.

4. How is Mitsue Kanko governed, and what's the future relationship? Who holds decision power? Does the plan assume their full agreement? Should the project plan be formulated autonomously?

Governance of Mitsue Kanko itself: a member-owned 森林組合 under the Forestry Cooperative Act — forest-owning members (組合員) hold shares (出資金, ¥31.3M capital), elect a board (理事会) led by a representative director (代表理事組合長 — one source names Tanaka Toshihiro, unconfirmed on the public registry), with a general meeting of members as the top decision body. It belongs to the 奈良県森林組合連合会 (prefectural federation). Financials aren't public — a structural blind spot, not something either party is missing.

Relationship to the project entity: currently informal. The stakeholders doc lists "Local Forestry Group" as an **"Early consultation partner (met early 2026)"** — not a signed MOU, not a board seat, not a formal agreement. Mitsue Kanko is separately envisioned as the **sole operator** of planting/tending work.

Does the plan assume full agreement/collaboration? Yes. The business case treats "Forestry co-op / landowners" as a stakeholder who is *asked* for sugi harvesting access and restoration participation, and gets fair compensation + J-Credit income in return — via **"Phase-1 contracts"** that don't exist yet. Their formal buy-in is a Phase 1 dependency, not something already secured. The plan's entire workforce-scaling mechanism (Q1 above) depends on Mitsue Kanko agreeing to grow, mechanize, and be funded by an outside energy project — a real ask of their board/general-meeting, not a foregone conclusion.

Should the project plan be formulated autonomously? Drafting it solo is fine — someone has to write the first version. The overstep would be presenting it to the village, funders, or Mitsue Kanko as a settled arrangement rather than a **contingent proposal** pending the cooperative's own board/general-meeting approval. Recommend the documents keep using conditional framing ("proposed," "Phase 1 contracts pending") wherever they describe Mitsue Kanko's role, and that a direct approach to their 代表理事組合長 becomes an explicit Phase 0/1 action item if it isn't already one.